Theodore Rogalski

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Education

Stevens Institute of Technology, New Jersey **Expected May 2027**

Joint Bachelors & Masters of Engineering in Computer and Electrical Engineering *GPA: 3.93*

Relevant Coursework: Optimization, Autonomous Mobile Robotics, Probability and Stochastic Processes, Control Theory

Leadership and Activities: Sigma Phi Epsilon Vice President of the Learning Community, Teaching Assistant (Introduction to Programming and Algorithmic Thinking)

Honors: California Seal of Biliteracy (Spanish), Dean's List, Member: IEEE HKN, RAS, CSS, $500 NJ STEM Grant Winner

Experience

Robotics Summer School Intern, Los Alamos National Laboratory, *Los Alamos, NM* **May 2026 - August 2026**

- Developed a safe reinforcement learning-based control policy in IsaacSim/Lab for DNA sample transfer that enabled fully autonomous operation of an auto lab environment with no collisions observed during hours of testing and under 5 mm sim-to-real position error
- Created a computer vision system for accurate sim to real transfer of policy with 1% error rate during testing, allowing for execution in unpredictable operational lab environment
- Created Human Machine Interface for active monitoring of system with digital twin & fully autonomous operation
- Planned submission of findings to **IDETC 2027**

AI Research Intern, Air Force Research Lab, *Rome, NY* **June 2025 - August 2025**

- Trained and evaluated Vision Transformer, CNN and Multi-Modal LLM models with custom classifier layers on custom image dataset created using the Unreal Engine 3D environment
- Utilized PyTorch and CUDA GPU integration for parallel processing, achieving 10x speedup for training and inference with a 30% accuracy increase on inductive tasks
- Utilized traceability techniques and permutation analysis to increase AI transparency

Graduate Researcher, Stevens Institute of Technology, *Hoboken, NJ* **September 2026 - Present**

- Developed low-level controller for operation of an unmanned surface vehicle using PID control and machine learning
- Developed multi-agent reinforcement learning controller using computer vision for a ROS2, VRX simulation system optimized for low-power hardware requirements and custom URDF models
- Mathematically modeled and tuned controller for oceanic USV environment

Undergraduate Researcher, Stevens Undergraduate Research, *Hoboken, NJ* **April 2025 - August 2026**

- Created custom simulation environment for Multi-Agent Reinforcement Learning on the graph topology for persistent monitoring in Python using Pettingzoo, NetworkX
- Formulated a deep reinforcement learning algorithm orchestrating a neural network state estimator, the graph Laplacian, Graph Attention, an Actor-Critic algorithm using MLPs and a discrete control barrier function for multi-agent systems
- **Published in and presenting at the IEEE HKN Innovating the Future Conference, Nov. 2026**

Student Assistant, California Air Resources Board, *Sacramento, CA* **December 2023 - August 2024**

- Independently built Vehicle Identification Number (VIN) Decoder System using NHTSA VIN Decoder API for large scale data processing
- Developed internal data processing tool that saved hundreds of thousands of dollars for state agency
- Created a secure Graphical User Interface (GUI) in Python's Flask framework with 2FA and input sanitization

Projects

Reinforcement Learning-Based Buck-Boost Controller | *MATLAB, Simulink* **November - December 2025**

- Implemented a Simple Actor Critic Reinforcement Learning algorithm for PWM Control of a buck-boost system with limited system information, achieving steady state control within 10% of accuracy compared to a baseline PID controller
- Developed an autotuned PID controller for a baseline of the system

Cybersecurity Transceiver Module | *VHDL, FPGA, Graphics Programming* **January - May 2026**

- Created a SHA256 hashing system for secure message-sending with the ability for users to input messages to an FPGA on a monitor using keyboard peripherals, achieving millisecond speed

Technical Skills

Machine Learning: Pettingzoo, Gymnasium, PyTorch, Agent Orchestration (Langchain, Claude Code)
Languages/Database: C++, Python, SQL, JS
Hardware: Embedded C, Universal Robots Manipulators, Microcontrollers
Software Tools: Docker, Git, Solidworks, AutoCAD, Matlab, R, ARM Assembly, Local Networking